UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

Ideas for exam projects

Course of:
Statistics and data analysis

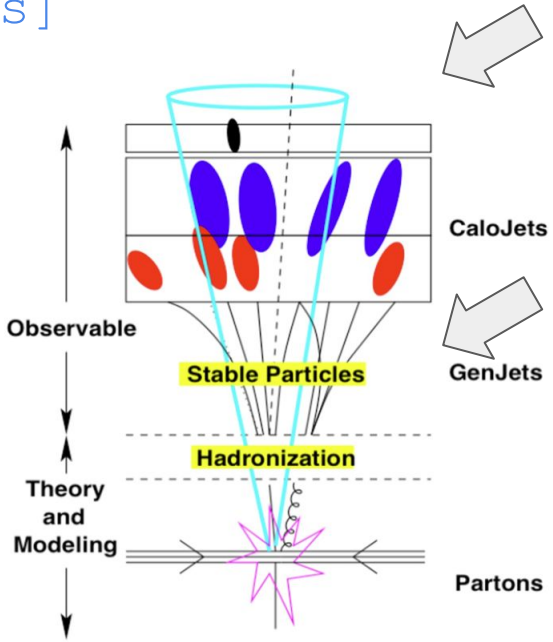
Federico De Guio - Martino Borsato

Unfold all observables with OmniFold - [HEP physics]

- **Jet substructure** refers to a set of theoretical and experimental techniques used in particle physics to resolve and analyze the internal radiation pattern of a jet
  - Allows physicists to distinguish between standard QCD background and boosted heavy particles like top quarks, W/Z bosons, or the Higgs boson

- Starting from the original OmniFold paper, replicate the results described by **unfolding the detector response on 6 jet substructure observables**

- Compare and discuss the results obtained against the Iterative Bayesian Unfolding method
  - Which one does better?
  - How expensive is OmniFold with respect to IBU?



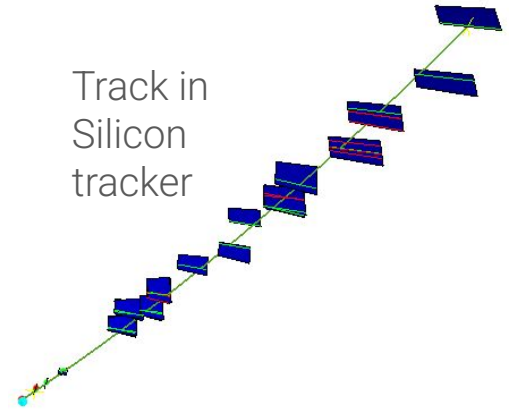
- The DATASET is available on zenodo and can be accessed directly using the EnergyFlow package

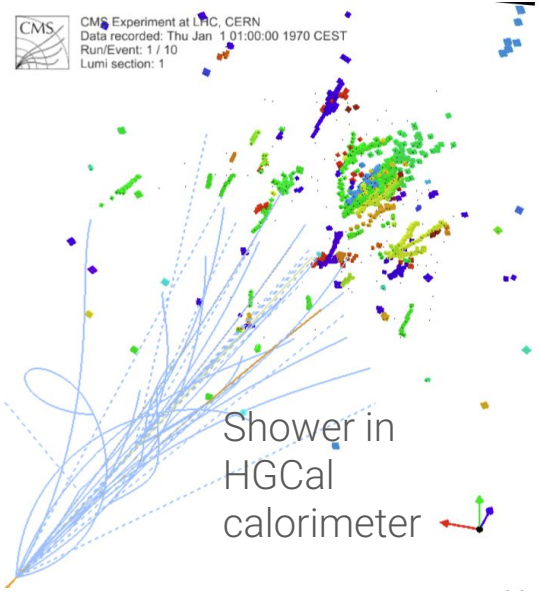
Particle Tracking with a GNN - [HEP physics]

- The reconstruction of charged **particle trajectories** in tracking detectors and **particle showers** in calorimeters are challenging tasks with common characteristics:
  - **high dimensionality**
  - **high degree of sparsity**
  - **complex geometric layouts**

- **Graph Neural Networks** are the perfect tool to perform edge classification to build tracks or reconstruct calorimetric showers
  - Edge classifier to reject any edge that doesn't connect points that belong to the same particle

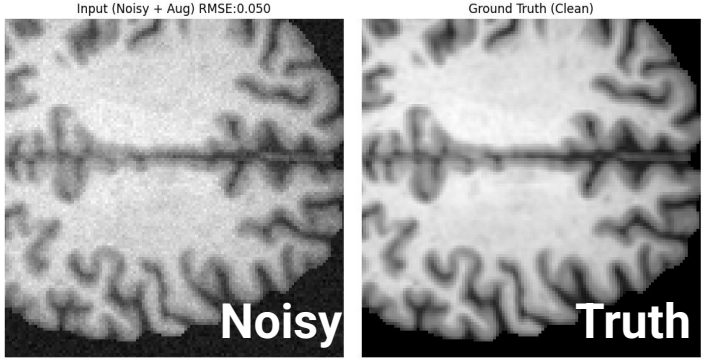
- Follow the path of this paper and **implement a GNN** to reconstruct the tracks in events from the dataset available on Kaggle here
  - Demonstrate the ability to cluster tracker hits together





Denoising of brain images from MRI - `[Medical physics]`

- Demonstrating the ability of **reducing noise on MRI images** during post-processing allows segmentation and classification algorithms to **better identifying pathologies**. If noise can be handled and subtracted, a **faster acquisition** of the image is also possible



- Implement a ML algorithm to **perform denoising** on a dataset of brain images acquired using a RMI scanner

- A dataset is available where images affected by different noise levels are collected:
  - `_small` → low gaussian noise
  - `_high` → high gaussian noise
  - `_mixed` → different levels of noise
  - `_true` → image without noise

- Different noise levels can be used to study the **ability of the algo to generalize the task**

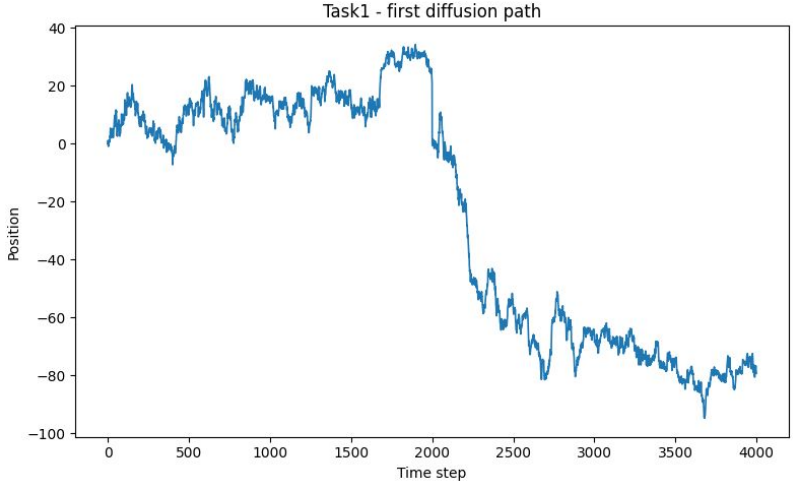
Decode anomalous diffusion - [Bio-physics]

- Decode diffusion is **crucial parameter to understand how life works at the molecular level** (indicator of the cell health)

- In classical diffusion (**Brownian motion**) the Mean Squared Displacement is proportional to time (alfa=1)

$$MSD \propto t^{\alpha}$$

  - Alfa < 1 → more common and due to sub-diffusion (e.g. trapping, macromolecular crowding, viscoelasticity)



- With reference to the paper "Objective comparison of methods to decode anomalous diffusion", analyze the time sequences of this dataset and implement supervised ML algos to:
  - Task1: Choose the best model to **infer the alfa parameter** → sub- vs classic vs superdiffusion
  - Task2: **Classify diffusion model**

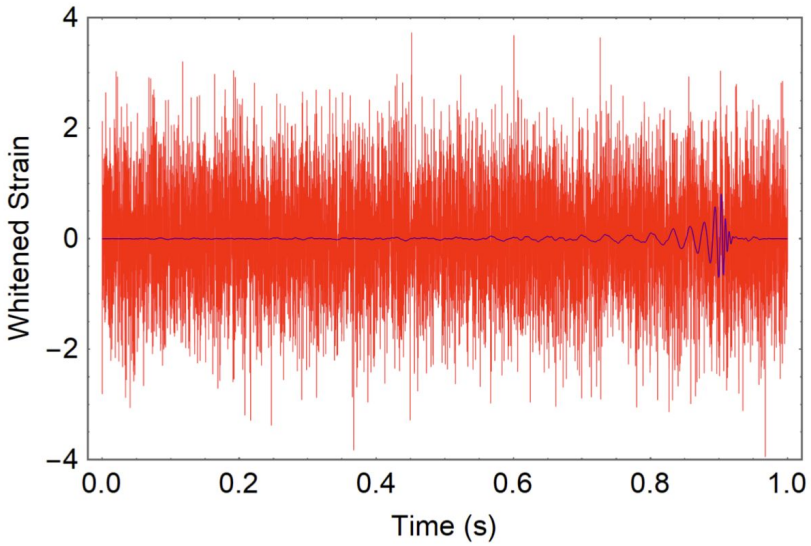
GW signals - [Astrophysics]

- Since 2015, **gravitational waves have provided a new lens to listen to the universe**
  - Experiments such as LIGO or VIRGO report observations of events like the merge of binary systems

- Using the Effective-one-body model, **generate a library of black-hole binaries** with realistic noise levels

- Following the approach described in this paper build **one or more models** to:
  - **Identify signals** of mergers buried in the noise
  - **Predict the parameters** of the binary system

Sample signal injected into real LIGO noise



Measure neutrino mixing with Daya Bay data [particle physics]

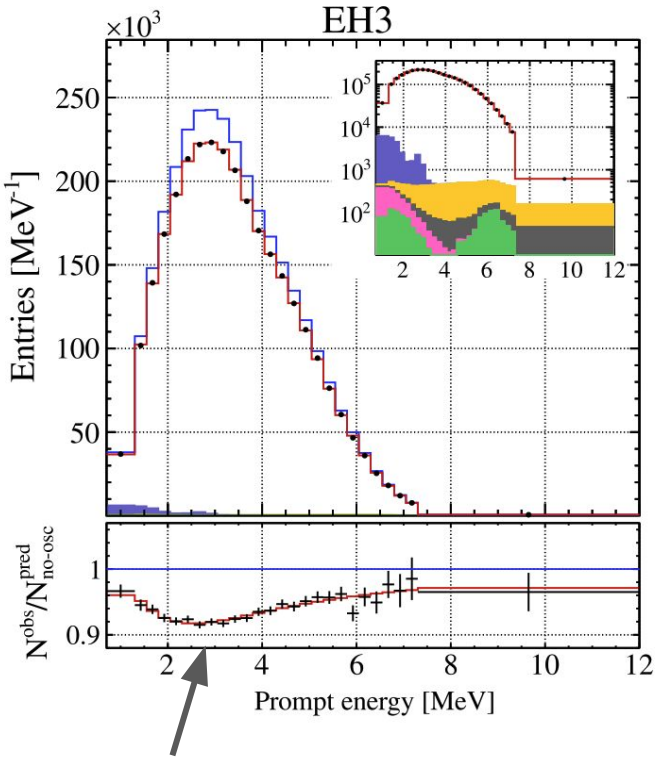
PhysRevLett.130.161802

Daya Bay Reactor Neutrino Experiment

- **Goal:** measure the neutrino mixing angle θ_{13} using electron antineutrinos from nuclear reactors
- Neutrinos detected via inverse beta decay

$$\bar{\nu}_e + p \rightarrow e^+ + n$$

- Eight detectors at 400-1800 m distance from reactors
- Near detectors monitor flux, far detectors observe disappearance due to neutrino fluctuation



Neutrino disappearance due to fluctuation

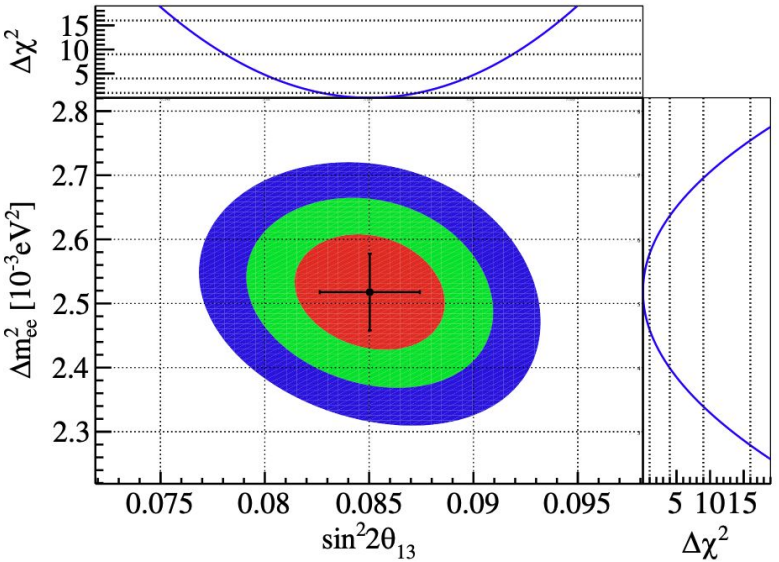
Measure neutrino mixing with Daya Bay data [particle physics]

Use the "analysis dataset" from zenodo
https://zenodo.org/records/17587229

The **dayabay-model python package** provides the predicted IBD spectra and the chi2 function

Can build chi2, fit and extract parameter values and **confidence intervals** while profiling over nuisance parameters
You could also set up a **Bayesian** analysis!

Important **nuisance parameter**: reactor flux is not known but it's determined from data (mostly from near-hall detectors which can measure it before oscillation)



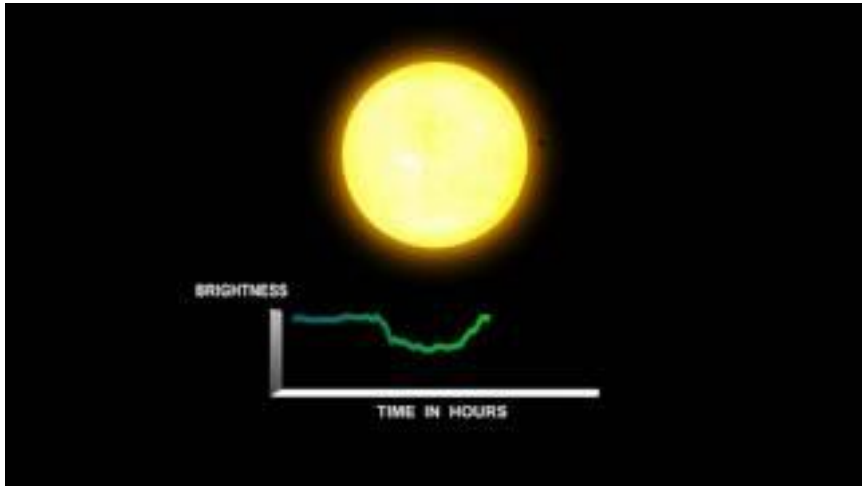
Exoplanet Hunting in Deep Space [astrophysics]

Exoplanets in deep space can be identified by tiny periodic transit dips in star brightness

- **Kepler/TESS** spacecrafts (2009/2018) monitored 150k stars every 30' for 4 years
- **Dataset**: NASA MAST Archive — raw Kepler/TESS light curves; also pre-packaged on Kaggle
- Training set with 37 confirmed exoplanets and 5k non-exoplanets (unbalanced)

Tasks:

- Classification to automate detection
- Estimate transit depth and duration with uncertainty
- Can infer planet radius R_p using star radius $R_\star$ taken from astroquery



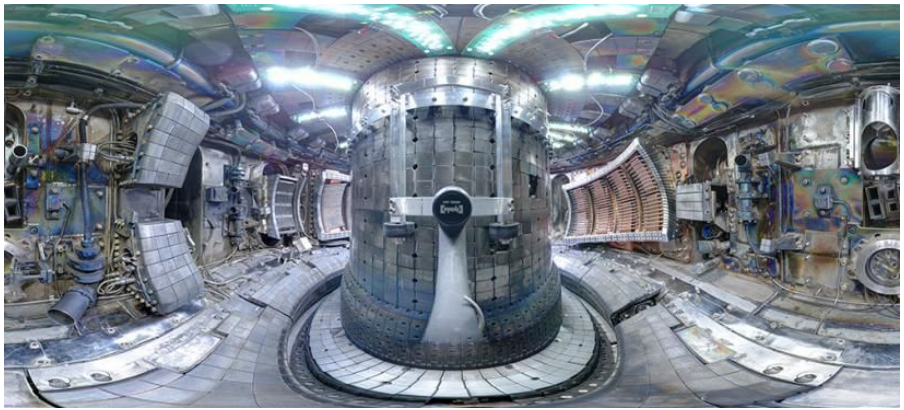
$$R_p = R_\star \sqrt{\delta}$$

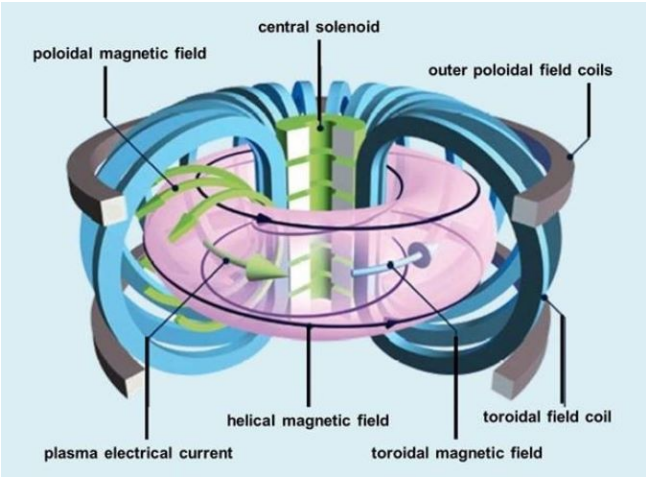
Plasma Physics & Fusion Energy (Density limit instability)

Maris, A. D., Rea, C., Trevisan, G. L., & the Alcator C-Mod Team. The Open Density Limit Database. MIT Plasma Science and Fusion Center (2025).

- Dataset with measurements from the Alcator C-Mod tokamak
- **Density limit:** the plasma collapses when its density gets too high. Challenge for sustained fusion reactors
- **Dataset**: each row contains a single time point in a discharge with plasma parameters and density limit phase label.
- Goal: classify whether the shot ended in a density-limit disruption or not and identify which features drive the prediction

For more plasma physics project, Prof Cavedon is available for a chat

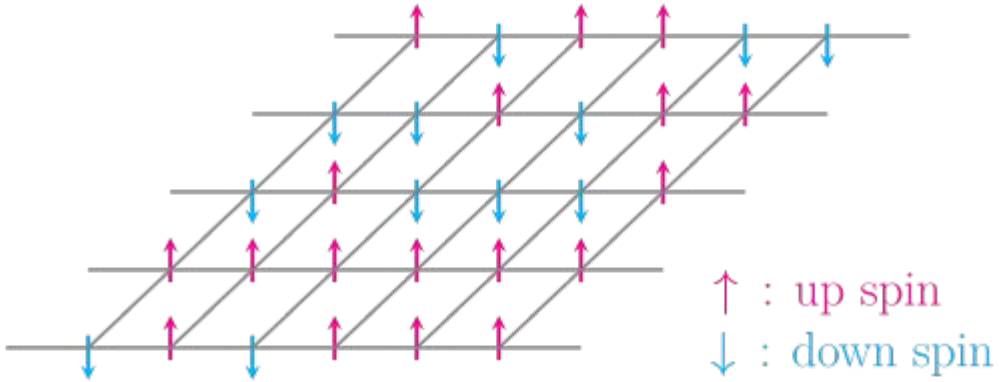




Statistical Mechanics & Criticality (Unsupervised Learning)

Reference paper: *A high-bias, low-variance introduction to Machine Learning for physicists* (Mehta et al., 2019)

- Use the dataset at this link
- 160k spring configurations with Lattice size 40x40 and various temperatures
- Information on the phase is encoded in long-rance correlations between spins in the lattice.



The Task: Feed the unlabeled spin configurations into an **unsupervised Autoencoder**.

By analyzing the low-dimensional latent space, show how the network **automatically discovers the order parameter** (magnetization) and cleanly identifies the critical phase transition temperature Tc.